

Literature Research Working Document

Effects of arable land measures on carabids and earthworms

Field	Entry
Prepared by	Lokesh Goud Dabbi
Supervisor	Dr. Katrin Ronnenberg
Topic	Effects of agri environmental measures on arable land on carabid beetles and earthworms
Databases	Web of Science Core Collection; Scopus
Date of searches	04.08.2026

★ Progress at a glance

Step	Task	Status
1	Compare Web of Science and Scopus; copy source lists; compare hits	Done
2	Develop the search syntax and save it with the date	Done
3	Reduce to a manageable number of papers	Done (178 carabid, 37 earthworm)
4	Rank papers by relevance from the abstracts (score 1 to 3)	Done
5	Write excerpts of the most relevant papers	Done (Section 7)

1 Research question for better understanding

How do agri environmental measures on arable land, such as flower strips, beetle banks, field margins, buffer strips, set aside and conservation headlands, affect the abundance and diversity of carabid beetles?

Earthworms and litter are treated as separate response variables in a second search (Search B), as agreed with the supervisor.

2 Concept table

The rows are types of term and the columns are the three concepts. Terms within a column are combined with OR, and the columns are combined with AND.

	Organism	Response	System and measure
Subject	Carabid*	Abundance and density	Arable land and measures
Synonyms	Carabidae	density, catch rate	cropland, farmland
Related terms		cover of plants, vegetation density, litter	crop rotation
Broader terms	beetles, Coleoptera	community metrics	agri environmental measures
Narrower terms	Carabus auratus, C. cancellatus, Pterostichus, Poecilus	species richness, diversity	beetle banks, flower strips, field margins, set aside, conservation headlands

3 Web of Science and Scopus (Task 1)

The source information was taken from clarivate.com (Web of Science Core Collection) and elsevier.com/products/scopus/content, checked on 04.08.2026.

Aspect	Web of Science (Clarivate)	Scopus (Elsevier)
Structure	A platform of several databases; the flagship is the Core Collection	A single integrated database

Indexes	SCI Expanded, SSCI, A&HCI, ESCI, Conference Proceedings, Book Citation and others	Not divided into indexes
Journals	More than 22,000 covered in full	More than 25,000 active, from over 5,000 publishers
Books	Book Citation Index (from 2005)	About 237,000 non serial books and over 63,000 series titles
Conferences	Conference Proceedings Citation Index (from 1990)	About 10.5 million conference papers
Coverage depth	Back to 1900 in SCI Expanded	About 83 million records, in depth from 1996 and older records to the 1800s
Extra databases	BIOSIS, Zoological Record, CAB Abstracts, MEDLINE	Patent links (about 46 million)
Selection body	Clarivate editorial team	Content Selection and Advisory Board

4 Search syntax (Task 2)

In Web of Science the strings were entered in Fielded Search with the field set to Topic (no TS= prefix). In Scopus they were entered in Advanced document search using TITLE ABS KEY. All searches were run on 04.08.2026.

Search A, Carabids (Web of Science)

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(carabid* OR Carabus OR Pterostichus OR Poecilus)
AND (abundance OR density OR richness OR biomass)
AND (arable OR cropland OR farmland OR cereal* OR crop*)
AND ("flower strip" OR "flower strips" OR "beetle bank" OR "beetle banks"
OR "field margin" OR "field margins" OR "buffer strip" OR "buffer strips"
OR "conservation headland" OR "conservation headlands" OR set-aside
OR "sown margin" OR "grass margin")
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Search A, Carabids (Scopus)

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TITLE-ABS-KEY((carabid* OR Carabus OR Pterostichus OR Poecilus)
AND (abundance OR density OR richness OR biomass)
AND (arable OR cropland OR farmland OR cereal* OR crop*)
AND ("flower strip" OR "flower strips" OR "beetle bank" OR "beetle banks"
OR "field margin" OR "field margins" OR "buffer strip" OR "buffer strips"
OR "conservation headland" OR "conservation headlands" OR set-aside
OR "sown margin" OR "grass margin"))
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Search B, Earthworms and litter (Web of Science)

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(earthworm* OR Lumbric* OR Lumbricidae)
AND (abundance OR density OR biomass OR litter OR "leaf litter"
OR "litter cover" OR "litter decomposition" OR "litter mass")
AND (arable OR cropland OR farmland OR cereal* OR crop*)
AND ("flower strip" OR "flower strips" OR "beetle bank" OR "beetle banks"
OR "field margin" OR "field margins" OR "buffer strip" OR "buffer strips"
OR "conservation headland" OR "conservation headlands" OR set-aside
OR "sown margin" OR "grass margin")
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Search B, Earthworms and litter (Scopus)

```
TITLE-ABS-KEY((earthworm* OR Lumbric* OR Lumbricidae)
AND (abundance OR density OR biomass OR litter OR "leaf litter"
OR "litter cover" OR "litter decomposition" OR "litter mass")
AND (arable OR cropland OR farmland OR cereal* OR crop*)
AND ("flower strip" OR "flower strips" OR "beetle bank" OR "beetle banks"
OR "field margin" OR "field margins" OR "buffer strip" OR "buffer strips"
OR "conservation headland" OR "conservation headlands" OR set-aside
OR "sown margin" OR "grass margin"))
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Note: in Web of Science the Topic field is set by the dropdown, so the TS= prefix is not typed. In Scopus the query must open with TITLE ABS KEY((and close with)). Inside quoted phrases the wildcard is unreliable, so singular and plural forms are written out in full.

5 Hit counts and de duplication (Task 3)

No.	Date	Database	Search	Hits	Comment
1	04.08.2026	WoS Core Coll.	A, broad terms	277	Before the measure terms
2	04.08.2026	Scopus	A, broad terms	163	Before the measure terms
3	04.08.2026	WoS Core Coll.	A, with measures	161	Exported to RIS
4	04.08.2026	Scopus	A, with measures	113	Exported to RIS
5	04.08.2026	WoS and Scopus	A, merged	178	Unique after de duplication
6	04.08.2026	WoS and Scopus	B, merged	37	Unique after de duplication

The results were exported and imported into Zotero and de duplicated. The final sets to screen were 178 carabid papers from Search A and 37 earthworm and litter papers from Search B.

6 Relevance ranking (Task 4)

Each paper was scored from 1 to 3 by reading only the title and abstract, with one score per paper. A score of 1 means the paper directly answers the question (the organism, arable land and a named measure all present); 2 means it is partly relevant; and 3 means it is background, a review or off topic.

Score	Meaning	Carabids	Earthworms
1 core	Directly answers the question	81	20
2 partial	Right taxon, wrong system, or right measure, other focus	48	5
3 background	Reviews, methods, general or off topic	32	6
no abstract	Could not be screened, as no abstract was available	17	6

7 Most relevant papers and excerpts (Task 5)

The full list of all 101 directly relevant (score 1) papers is given in the References in Section 9, together with their DOIs. From those, a smaller set of the most relevant papers was chosen for the short excerpts below. These are the papers with the clearest comparison between margins and fields and with reported effect sizes, chosen so that every measure type is represented (beetle banks, conservation headlands, flower and wildflower strips, buffer strips, field margins and set aside). Both positive results and papers that found no effect are included, so that the evidence is represented fairly.

7.1 Earthworms and litter, eight chosen papers

Margins and strips increase earthworm abundance

Pelosi et al. (2024) In northern France, 2.5 year old sown wildflower strips held far more earthworms and enchytraeids than the adjacent cropped fields, with abundance up by roughly 69 and 61 percent. Anecic and endogeic earthworms benefited most.

Crittenden et al. (2015) Across four Dutch farms over three years, field margin strips held far more earthworms than the ploughed fields beside them (about 284 against 164 per square metre, and biomass 84 against 31 grams per square metre).

Nieminen et al. (2011) At 53 Finnish sites, field margins held over twice the earthworm density and nearly double the number of species of the adjacent cultivated fields, and richness inside the fields fell with distance from the margin.

Li et al. (2022) In a rape and rice system, widening the field margin made a large difference: during post harvest flooding, earthworm abundance in the widened margins was some sixteen to twenty times that of the conventional margin, and wider margins let the field recover faster.

Roarty and Schmidt (2013) In five Irish arable fields over three years, both permanent and newly created margin strips held many more earthworms than the field interior (roughly 470 and 371 against 155 per square metre), although no spillover into the field itself was found.

Refuge, spatial and no effect findings

Prendergast Miller et al. (2021) Grass clover ley strips introduced into four arable fields reached about 732 earthworms per square metre within two years, around four times the arable soil. The increase came from the field's own residual population rather than from the margins.

Frazao et al. (2017) In a Dutch polder, earthworm density, biomass and richness did not differ significantly between field margins and arable fields, a useful null result; only community composition differed.

Kiss et al. (2019) In two UK fields, frequent flooding cut earthworms sharply in the crop (down about 59 percent) but actually raised them in the margins (up about 141 percent), so the margins buffered earthworms against flooding stress.

7.2 Carabid beetles, ten chosen papers

Beetle banks and conservation headlands

Collins et al. (2002) In winter wheat, excluding the predators that spread out from a beetle bank let aphid numbers rise by about a third, while carabid catches fell by roughly a third to a half. This is direct evidence of pest control spilling from the bank into the crop.

Hassall et al. (1992) In the Breckland ESA, carabid catches were higher in conservation and uncropped headlands than in fully sprayed ones, and higher again in the crop next to uncropped headlands.

Flower strips and wildflower strips

Zhang et al. (2025) A meta analysis of flower strips found a clear positive effect on the richness of ground active predators, and narrow linear strips were about four times more effective than large flower areas.

Mei et al. (2021) Across sixteen Dutch wheat fields, simply having a wildflower strip made little difference on its own; what mattered was the cover and richness of flowers, which drove increases in carabids and spiders. Effects inside the crop reached only a few metres.

Ganser et al. (2019) Carabids overwintered in greater numbers in wildflower strips than in wheat, but ploughing annual strips over winter cut emergence by about two thirds, so a poorly managed strip can act as an ecological trap.

Field margins and buffer strips

Bennewicz and Barczak (2020) In Poland, field margins held more abundant and more diverse carabids than the arable fields, including *Carabus auratus*, and the bushy margins were the richest refuge and held protected species.

Liu et al. (2006) On the North China Plain, field margins held significantly more carabid species and individuals than the species poor field interiors, confirming their role as refuges.

Burmeister et al. (2025) Across forty Bavarian fields, buffer strips did not raise local carabid diversity (unbuffered borders sometimes held more), but they did support more distinct assemblages, adding to diversity between sites.

Feehan et al. (2005) An assessment of the Irish REPS scheme found few differences in carabid richness or abundance between scheme and non scheme farms, a null result, with variation driven mostly by hedge age and gappiness.

Collins et al. (2003) Comparing five grasses sown on beetle banks, *Arrhenatherum elatius* and *Dactylis glomerata* supported the highest overwintering predator densities, offering practical guidance for building banks.

8 Note on unscored records

Twenty three records, seventeen carabid and six earthworm or litter, contained no abstract and so could not be screened at the abstract stage. They remain unscored, and their abstracts would need to be retrieved from the publishers before they could be included.

9 Full references, all score 1 papers with DOIs

The complete list of the 101 directly relevant papers, in alphabetical order and with DOI links. This list also serves as the full record of the score 1 papers.

A. Earthworms and litter, 20 papers

- Cameron, A., Boilard, G., Dubois, R., Bradley, R., Benetková, P., Józefowska, A. et al. (2021). Distribution of earthworm communities in agroecosystems with forested riparian buffer strips: A multiscale study. *Applied Soil Ecology*, 167. <https://doi.org/10.1016/j.apsoil.2021.104035>
- Crittenden, S., Huerta, E., de Goede, R. and Pulleman, M. (2015). Earthworm assemblages as affected by field margin strips and tillage intensity: An on farm approach. *European Journal of Soil Biology*, 66, 49-56. <https://doi.org/10.1016/j.ejsobi.2014.11.007>
- Fraza, J., de Goede, R., Brussaard, L., Faber, J., Groot, J. and Pulleman, M. (2017). Earthworm communities in arable fields and restored field margins, as related to management practices and surrounding landscape diversity. *Agriculture Ecosystems & Environment*, 248, 1-8. <https://doi.org/10.1016/j.agee.2017.07.014>
- Gormsen, D., Hedlund, K., Korthals, G., Mortimer, S., Pizl, V., Smilauerova, M. et al. (2004). Management of plant communities on set aside land and its effects on earthworm communities. *European Journal of Soil Biology*, 40(3-4), 123-128. <https://doi.org/10.1016/j.ejsobi.2004.08.001>
- Hof, A. and Bright, P. (2010). The impact of grassy field margins on macro invertebrate abundance in adjacent arable fields. *Agriculture Ecosystems & Environment*, 139(1-2), 280-283. <https://doi.org/10.1016/j.agee.2010.08.014>
- Holden, J., Grayson, R., Berdeni, D., Bird, S., Chapman, P., Edmondson, J. et al. (2019). The role of hedgerows in soil functioning within agricultural landscapes. *Agriculture Ecosystems & Environment*, 273, 1-12. <https://doi.org/10.1016/j.agee.2018.11.027>
- Kernecker, M., Whalen, J. and Bradley, R. (2015). Agricultural management and flooding shape habitats for non native earthworms in southern Quebec, Canada. *Applied Soil Ecology*, 96, 240-250. <https://doi.org/10.1016/j.apsoil.2015.08.011>
- Kiss, T., Chen, X., Ponting, J., Sizmur, T. and Hodson, M. (2021). Dual stresses of flooding and agricultural land use reduce earthworm populations more than the individual stressors. *Science of the Total Environment*, 754. <https://doi.org/10.1016/j.scitotenv.2020.142102>
- Lavergne, S., Halde, C. and Lynch, D. (2025). Earthworm abundance and diversity in response to intensive crop management in organic field crop farms of southern Quebec, Canada. *Applied Soil Ecology*, 206. <https://doi.org/10.1016/j.apsoil.2024.105850>
- Li, C., Zhao, Y., Cheng, K., Zhou, J., Xiao, X., Iqbal, A. et al. (2022). Expansion of Field Margin Size Mitigate the Hazard of Rotary Tillage to Earthworm in Rice Rape Rotation System. *Agronomy Basel*, 12(9). <https://doi.org/10.3390/agronomy12092209>
- Mittmannsgruber, M., Monoshyn, D., Slawiczek, C., Taheri, K., Wollenweber, F., Meissner, B. et al. (2025). Earthworm and soil microarthropod abundances affected by agricultural land use and soil properties. *Pedobiologia*, 112. <https://doi.org/10.1016/j.pedobi.2025.151073>

- Mittmannsgruber, M., Monoshyn, D., Gruber, E., Wiedenegger, E., Csuzdi, C., Pascher, K. et al. (2026). From cropland to alpine meadows: Earthworm diversity in agroecosystems across Austria. *Agriculture Ecosystems & Environment*, 402. <https://doi.org/10.1016/j.agee.2026.110312>
- Nieminen, M., Ketoja, E., Mikola, J., Terhivuo, J., Sirén, T. and Nuutinen, V. (2011). Local land use effects and regional environmental limits on earthworm communities in Finnish arable landscapes. *Ecological Applications*, 21(8), 3162–3177. <https://doi.org/10.1890/10-1801.1>
- Nuutinen, V., Butt, K. and Jauhiainen, L. (2011). Field margins and management affect settlement and spread of an introduced dew worm (*Lumbricus terrestris* L.) population. *Pedobiologia*, 54, S167-S172. <https://doi.org/10.1016/j.pedobi.2011.07.010>
- Pelosi, C., Bertrand, M. and Gardarin, A. (2024). Wildflower strips of 2.5 year old promote earthworms and enchytraeids (Annelida, Oligochaeta) in arable fields. *European Journal of Soil Biology*, 122. <https://doi.org/10.1016/j.ejsobi.2024.103644>
- Prendergast-Miller, M., Jones, D., Berdeni, D., Bird, S., Chapman, P., Firbank, L. et al. (2021). Arable fields as potential reservoirs of biodiversity: Earthworm populations increase in new leys. *Science of the Total Environment*, 789. <https://doi.org/10.1016/j.scitotenv.2021.147880>
- Roarty, S. and Schmidt, O. (2013). Permanent and new arable field margins support large earthworm communities but do not increase in field populations. *Agriculture Ecosystems & Environment*, 170, 45-55. <https://doi.org/10.1016/j.agee.2013.02.011>
- Smith, J., Potts, S. and Eggleton, P. (2008). The value of sown grass margins for enhancing soil macrofaunal biodiversity in arable systems. *Agriculture Ecosystems & Environment*, 127(1-2), 119-125. <https://doi.org/10.1016/j.agee.2008.03.008>
- Smith, J., Potts, S., Woodcock, B. and Eggleton, P. (2008). Can arable field margins be managed to enhance their biodiversity, conservation and functional value for soil macrofauna?. *Journal of Applied Ecology*, 45(1), 269-278. <https://doi.org/10.1111/j.1365-2664.2007.01433.x>
- Woodcock, B., Cook, S., Hulmes, L., Hulmes, S., Torrance, M., Redhead, J. et al. (2025). Agroecological farming promotes yield and biodiversity but may require subsidy to be profitable. *Journal of Applied Ecology*, 62(8), 1902-1913. <https://doi.org/10.1111/1365-2664.70079>

B. Carabid beetles, 81 papers

- Al Hassan, D., Georgelin, E., Delattre, T., Burel, F., Plantegenest, M., Kindlmann, P. et al. (2013). Does the presence of grassy strips and landscape grain affect the spatial distribution of aphids and their carabid predators?. *Agricultural and Forest Entomology*, 15(1), 24-33. <https://doi.org/10.1111/j.1461-9563.2012.00587.x>
- Alignier, A. and Aviron, S. (2017). Time lagged response of carabid species richness and composition to past management practices and landscape context of semi natural field margins. *Journal of Environmental Management*, 204, 282-290. <https://doi.org/10.1016/j.jenvman.2017.08.054>
- Anjum-Zubair, M., Entling, M., Bruckner, A., Drapela, T. and Frank, T. (2015). Differentiation of spring carabid beetle assemblages between semi natural habitats and adjoining winter wheat. *Agricultural and Forest Entomology*, 17(4), 355-365. <https://doi.org/10.1111/afe.12115>
- Bannwart, P., Gardarin, A. and Petit, S. (2025). Do semi natural habitats enhance overwintering of generalist predators in arable cropping systems? A meta analysis. *Biological Control*, 201. <https://doi.org/10.1016/j.biocontrol.2025.105700>
- Bannwart, P., Gardarin, A., Deytieux, V., Cordeau, S. and Petit, S. (2026). Pesticide free fields under minimum tillage and flower strips enhance carabid beetles and spiders through increased overwintering and spill over processes. *Agriculture Ecosystems & Environment*, 400. <https://doi.org/10.1016/j.agee.2026.110250>
- Barascou, L., Doehler, M., Buzy, S., Arnoult, A., Salembier, E., Richard, V. et al. (2025). Quantification of the biological control of aphids by their natural enemies in sugar beet crops. *Biological Control*, 205. <https://doi.org/10.1016/j.biocontrol.2025.105773>
- Bennewicz, J. and Barczak, T. (2020). Ground beetles (Carabidae) of field margin habitats. *Biologia*, 75(10), 1631-1641. <https://doi.org/10.2478/s11756-020-00424-y>
- Birkhofer, K., Wolters, V. and Diekötter, T. (2014). Grassy margins along organically managed cereal fields foster trait diversity and taxonomic distinctness of arthropod communities. *Insect Conservation and Diversity*, 7(3), 274-287. <https://doi.org/10.1111/icad.12051>

- Birkhofer, K., Fevrier, V., Heinrich, A., Rink, K. and Smith, H. (2018). The contribution of CAP greening measures to conservation biological control at two spatial scales. *Agriculture Ecosystems & Environment*, 255, 84-94. <https://doi.org/10.1016/j.agee.2017.12.026>
- Boetzel, F., Krimmer, E., Holzschuh, A., Krauss, J. and Steffan-Dewenter, I. (2022). Arthropod overwintering in agri environmental scheme flowering fields differs among pollinators and natural enemies. *Agriculture Ecosystems & Environment*, 330. <https://doi.org/10.1016/j.agee.2022.107890>
- Burmeister, J., Birnbeck, S., Panassiti, B., Blick, T. and Walter, R. (2025). Effect of buffer strips along small watercourses on farmland spiders (Araneae) and ground beetles (Carabidae). *European Journal of Entomology*, 122, 331-343. <https://doi.org/10.14411/eje.2025.038>
- CARDWELL, C., HASSALL, M. and WHITE, P. (1994). EFFECTS OF HEADLAND MANAGEMENT ON CARABID BEETLE COMMUNITIES IN BRECKLAND CEREAL FIELDS. *Pedobiologia*, 38(1), 50-62.
- Collins, K., Boatman, N., Wilcox, A., Holland, J. and Chaney, K. (2002). Influence of beetle banks on cereal, aphid predation in winter wheat. *Agriculture Ecosystems & Environment*, 93(1-3), 337-350. [https://doi.org/10.1016/S0167-8809\(01\)00340-1](https://doi.org/10.1016/S0167-8809(01)00340-1)
- Collins, K., Boatman, N., Wilcox, A. and Holland, J. (2003). A 5 year comparison of overwintering polyphagous predator densities within a beetle bank and two conventional hedgebanks. *Annals of Applied Biology*, 143(1), 63-71. <https://doi.org/10.1111/j.1744-7348.2003.00063.x>
- Collins, K., Boatman, N., Wilcox, A. and Holland, J. (2003). Effects of different grass treatments used to create overwintering habitat for predatory arthropods on arable farmland. *Agriculture Ecosystems & Environment*, 96(1-3), 59-67. [https://doi.org/10.1016/S0167-8809\(03\)00032-X](https://doi.org/10.1016/S0167-8809(03)00032-X)
- DENNIS, P. and FRY, G. (1992). FIELD MARGINS - CAN THEY ENHANCE NATURAL ENEMY POPULATION DENSITIES AND GENERAL ARTHROPOD DIVERSITY ON FARMLAND. *Agriculture Ecosystems & Environment*, 40(1-4), 95-115. [https://doi.org/10.1016/0167-8809\(92\)90086-Q](https://doi.org/10.1016/0167-8809(92)90086-Q)
- Dietzel, S., Batáry, P., Twerski, A., Kirmer, A., Kollmann, J., Tischew, S. et al. (2026). Synergistic effects of flower strips and landscape complexity buffer arthropods against warming and drought - A meta analysis. *Agriculture Ecosystems & Environment*, 411. <https://doi.org/10.1016/j.agee.2026.110617>
- Ditner, N., Balmer, O., Beck, J., Blick, T., Nagel, P. and Luka, H. (2013). Effects of experimentally planting non crop flowers into cabbage fields on the abundance and diversity of predators. *Biodiversity and Conservation*, 22(4), 1049-1061. <https://doi.org/10.1007/s10531-013-0469-5>
- Eyre, M., Labanowska-Bury, D., Avayanos, J., White, R. and Leifert, C. (2009). Ground beetles (Coleoptera, Carabidae) in an intensively managed vegetable crop landscape in eastern England. *Agriculture Ecosystems & Environment*, 131(3-4), 340-346. <https://doi.org/10.1016/j.agee.2009.02.006>
- Feehan, J., Gillmor, D. and Culleton, N. (2005). Effects of an agri environment scheme on farmland biodiversity in Ireland. *Agriculture Ecosystems & Environment*, 107(2-3), 275-286. <https://doi.org/10.1016/j.agee.2004.10.024>
- Fu, D., Zhu, Y., Duan, M., Yu, Z., Liu, Y. and Zhang, X. (2026). Distance effects of cropland boundaries on spider and carabid beetle assemblages in Beijing. *Journal of Economic Entomology*, 119(3), 1899-1910. <https://doi.org/10.1093/jee/toag106>
- Fusser, M., Holland, J., Jeanneret, P., Pfister, S., Entling, M. and Schirmel, J. (2018). Interactive effects of local and landscape factors on farmland carabids. *Agricultural and Forest Entomology*, 20(4), 549-557. <https://doi.org/10.1111/afe.12288>
- Gallé, R., Császár, P., Makra, T., Gallé-Szpisjak, N., Ladányi, Z., Torma, A. et al. (2018). Small scale agricultural landscapes promote spider and ground beetle densities by offering suitable overwintering sites. *Landscape Ecology*, 33(8), 1435-1446. <https://doi.org/10.1007/s10980-018-0677-1>
- Gallé, R., Geppert, C., Földesi, R., Tscharntke, T. and Batáry, P. (2020). Arthropod functional traits shaped by landscape scale field size, local agri environment schemes and edge effects. *Basic and Applied Ecology*, 48, 102-111. <https://doi.org/10.1016/j.baae.2020.09.006>
- Ganser, D., Knop, E. and Albrecht, M. (2019). Sown wildflower strips as overwintering habitat for arthropods: Effective measure or ecological trap?. *Agriculture Ecosystems & Environment*, 275, 123-131. <https://doi.org/10.1016/j.agee.2019.02.010>
- Geiger, F., Wäckers, F. and Bianchi, F. (2009). Hibernation of predatory arthropods in semi natural habitats. *Biocontrol*, 54(4), 529-535. <https://doi.org/10.1007/s10526-008-9206-5>

- Gonzalez, E., Strobl, M., Venturo, A., Parízek, V., Tajovsky, K. and Knapp, M. (2025). Wildflower strip establishment supports beneficial ground dwelling arthropods and pest control but has limited effects on weed seed control and spillover to adjacent fields. *Proceedings of the Royal Society B Biological Sciences*, 292(2048). <https://doi.org/10.1098/rspb.2025.0752>
- HASSALL, M., HAWTHORNE, A., MAUDSLEY, M., WHITE, P. and CARDWELL, C. (1992). EFFECTS OF HEADLAND MANAGEMENT ON INVERTEBRATE COMMUNITIES IN CEREAL FIELDS. *Agriculture Ecosystems & Environment*, 40(1-4), 155-178. [https://doi.org/10.1016/0167-8809\(92\)90090-X](https://doi.org/10.1016/0167-8809(92)90090-X)
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- Jowett, K., Milne, A., Garrett, D., Blumgart, D., Potts, S., Senapathi, D. et al. (2024). Using a multi scale approach to examine the effects of field margins and landscape features on predatory carabid communities in crop fields. *Agriculture Ecosystems & Environment*, 373. <https://doi.org/10.1016/j.agee.2024.109115>
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